MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Periodic Test Paper 10 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. Which term of the AP $3, 8, 13, 18, \ldots$ is 78? [1]

(A) 14th (B) 15th (C) 16th (D) 17th

Q2. Find the number of terms in the finite AP $7, 13, 19, \ldots, 205$: [1]

(A) 33 (B) 34 (C) 35 (D) 36

Q3. If the first term of an AP is 2 and the common difference is 4, then the sum of its first 40 terms is: [1]

(A) 3200 (B) 3280 (C) 3240 (D) 3300

Q4. The common difference of an AP whose n^{th} term is given by $a_n = 3n + 7$ is: [1]

(A) 3 (B) 7 (C) 10 (D) 4

Q5. **Assertion (A):** The sum of the first n odd natural numbers is always equal to n^2.
Reason (R): The first n odd positive integers form an AP with $a = 1$ and $d = 2$, whose sum is $S_n = \frac{n}{2}[2(1) + (n-1)2] = n^2$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. If the 14th term of an AP is twice its 8th term, and its 6th term is -8, find the first term and common difference. [2]

Q7. For what value of p are $2p - 1, 7, 3p$ three consecutive terms of an arithmetic progression? [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. How many three-digit natural numbers are completely divisible by 9? Also, find their total sum. [3]

Q9. The angles of a convex quadrilateral are in AP. If the greatest angle is double the least angle, find the measure of all four angles. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. The houses of a row in a colony are numbered consecutively from 1 to 49. Show that there exists a value of x such that the sum of the numbers of the houses preceding the house numbered x is equal to the sum of the numbers of the houses following it. Find this unique value of x. [5]

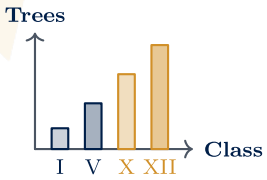
Q11. If the pth, q^{th}, and r^{th} terms of an AP are a, b, c respectively, prove that:

$$a(q-r) + b(r-p) + c(p-q) = 0$$

[5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Eco-Club School Tree Plantation Campaign



To inculcate environmental consciousness and reduce pollution, the Eco-Club of a senior secondary school plans a tree plantation drive along the perimeter boundary wall.
It is decided that the number of trees that each section of each class will plant will be identical to the class number (e.g., each section of Class I plants 1 tree, Class II plants 2 trees, and so on up to Class XII). Each class from Class I to XII has 3 equal sections.

Based on the above environmental project, answer the following questions:

(i) How many trees are planted by Class I (all 3 sections combined)? [1]

(ii) How many trees are planted by Class XII (all 3 sections combined)? [1]

(iii) Calculate the total number of trees planted by the entire school from Class I to Class XII. [3]

OR

(iv) If only the primary classes (Class I to Class V) are considered, calculate the total number of trees planted by them. [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★